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author: Aubrey Monaco
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```

Introduction

With warming temperatures due to anthropogenic climate change, bird migration times are shifting, and they are tending to migrate back to their breeding grounds earlier each year. Flowers that certain migrating birds, like Ruby-throated Hummingbirds (*Archilochus colubris*), feed on may be changing the time of their first bloom. I want to investigate how a pollinating bird species, the Ruby-throated Hummingbird, has been shifting its migration patterns and how the appearances of species it prefers to feed on, Red Buckeye (*Aesculus pavia*) and columbine (*Aquilegia canadensis*), has changed (or not changed) over the past 10 years. I predict that Ruby-throated Hummingbirds are migrating back earlier each year, and they are arriving at higher latitudes increasingly earlier than lower latitudes. Additionally, I suppose that the first sightings of these flowers in the months that Ruby-throated Hummingbirds migrate back North will be earlier compared to previous years.

Materials and methods

1. I installed leaflet, for making maps, and rinat and rnnpn to download data from iNaturalist and the US National Phenology Network, respectively. I then loaded my necessary libraries.

```
```{r, message=FALSE, warning=FALSE}
#install.packages("leaflet")
#install.packages("rinat")
#install.packages("rnnpn")
library(leaflet)
library(dplyr)
library(lubridate)
library(rinat)
library(rnnpn)
```
```

2. I loaded my data from iNaturalist and the US National Phenology Network and joined my data into one set.

- I repeated the below process of downloading observational for each year from 2001 to 2025 (no data was available from 2000), saving each to an object, and combining them to have a dataset for each year.

```
-rb = Red Buckeye
-cb = Columbine
```

```
```{r, message=FALSE, warning=FALSE}

#rb2001 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2001, place_id = 1, maxresults =
5000, quality = "research")
#cb2001 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2001, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2001 <- rbind(rb2001, cb2001)
```

```{r, message=FALSE, warning=FALSE, echo=FALSE}
#rb2002 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2002, place_id = 1, maxresults =
5000, quality = "research")
#cb2002 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2002, place_id = 1, maxresults
= 5000, quality = "research")
```

```

#fl_2002 <- rbind(rb2002, cb2002)

#rb2003 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2003, place_id = 1, maxresults =
5000, quality = "research")
#cb2003 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2003, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2003 <- rbind(rb2003, cb2003)

#rb2004 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2004, place_id = 1, maxresults =
5000, quality = "research")
#cb2004 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2004, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2004 <- rbind(rb2004, cb2004)

#rb2005 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2005, place_id = 1, maxresults =
5000, quality = "research")
#cb2005 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2005, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2005 <- rbind(rb2005, cb2005)

#rb2006 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2006, place_id = 1, maxresults =
5000, quality = "research")
#cb2006 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2006, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2006 <- rbind(rb2006, cb2006)

#rb2007 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2007, place_id = 1, maxresults =
5000, quality = "research")
#cb2007 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2007, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2007 <- rbind(rb2007, cb2007)

#rb2008 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2008, place_id = 1, maxresults =
5000, quality = "research")
#cb2008 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2008, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2008 <- rbind(rb2008, cb2008)

#rb2009 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2009, place_id = 1, maxresults =
5000, quality = "research")
#cb2009 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2009, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2009 <- rbind(rb2009, cb2009)

#rb2010 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2010, place_id = 1, maxresults =
5000, quality = "research")
#cb2010 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2010, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2010 <- rbind(rb2010, cb2010)

#rb2011 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2011, place_id = 1, maxresults =
5000, quality = "research")
#cb2011 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2011, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2011 <- rbind(rb2011, cb2011)

```

```
#rb2012 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2012, place_id = 1, maxresults =
5000, quality = "research")
#cb2012 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2012, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2012 <- rbind(rb2012, cb2012)

#rb2013 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2013, place_id = 1, maxresults =
5000, quality = "research")
#cb2013 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2013, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2013 <- rbind(rb2013, cb2013)

#rb2014 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2014, place_id = 1, maxresults =
5000, quality = "research")
#cb2014 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2014, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2014 <- rbind(rb2014, cb2014)

#rb2015 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2015, place_id = 1, maxresults =
5000, quality = "research")
#cb2015 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2015, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2015 <- rbind(rb2015, cb2015)

#rb2016 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2016, place_id = 1, maxresults =
5000, quality = "research")
#cb2016 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2016, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2016 <- rbind(rb2016, cb2016)

#rb2017 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2017, place_id = 1, maxresults =
5000, quality = "research")
#cb2017 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2017, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2017 <- rbind(rb2017, cb2017)

#rb2018 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2018, place_id = 1, maxresults =
5000, quality = "research")
#cb2018 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2018, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2018 <- rbind(rb2018, cb2018)

#rb2019 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2019, place_id = 1, maxresults =
5000, quality = "research")
#cb2019 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2019, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2019 <- rbind(rb2019, cb2019)

#rb2020 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2020, place_id = 1, maxresults =
5000, quality = "research")
#cb2020 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2020, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2020 <- rbind(rb2020, cb2020)

#rb2021 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2021, place_id = 1, maxresults =
```

```

5000, quality = "research")
#cb2021 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2021, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2021 <- rbind(rb2021, cb2021)

#rb2022 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2022, place_id = 1, maxresults =
5000, quality = "research")
#cb2022 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2022, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2022 <- rbind(rb2022, cb2022)

#rb2023 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2023, place_id = 1, maxresults =
5000, quality = "research")
#cb2023 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2023, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2023 <- rbind(rb2023, cb2023)

#rb2024 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2024, place_id = 1, maxresults =
5000, quality = "research")
#cb2024 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2024, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2024 <- rbind(rb2024, cb2024)

#rb2025 <- get_inat_obs(taxon_name = "Aesculus pavia", year = 2025, place_id = 1, maxresults =
5000, quality = "research")
#cb2025 <- get_inat_obs(taxon_name = "Aquilegia canadensis", year = 2025, place_id = 1, maxresults
= 5000, quality = "research")

#fl_2025 <- rbind(rb2025, cb2025)

```

3. I combined the datasets from each year into one. I then converted this data into a csv, and loaded it into my environment as nat\_flower to designate data from iNaturalist. I filtered nat\_flower to only include wild sightings of the selected species.

```

```{r, message=FALSE, warning=FALSE}
#fl_all <- rbind(fl_2001, fl_2002, fl_2003, fl_2004, fl_2005, fl_2006, fl_2007, fl_2008, fl_2009,
fl_2010, fl_2011, fl_2012, fl_2013, fl_2014, fl_2015, fl_2016, fl_2017, fl_2018, fl_2019, fl_2020,
fl_2021, fl_2022, fl_2023, fl_2024, fl_2025)

#write.csv(fl_all, "data/fl_all.csv")

nat_flower <- read.csv("data/fl_all.csv")
nat_flower <- nat_flower %>%
  filter(captive_cultivated == "false")
```

```

4. I then used the rnnp package to download data for the plant species from the National Phenology Network (which automatically saves as a csv). The species IDs were available through the NPN's website.

```

```{r, message=FALSE, warning=FALSE}
#Species IDs from USNPN

#nnp_download_individual_phenometrics(
# request_source='Aubrey Monaco',
# years=c(2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009,
2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023,2024, 2025),
# species_id=c(784, 149),
# download_path="nnp_flower.csv"
# )
```

```

5. I then combined my iNaturalist and NPN data into one dataset.

```

```{r}

```

```

nnp_flower <- read.csv("nnp_flower.csv")

all_flower <- nat_flower %>% full_join(nnp_flower)

```

```

...

```

6. I used the Lubridate package to standardize the format all of the dates in my flower data. Then, I used this standardized data to create a year column, where I pulled the years from the dates, and a year_day column, to transform the date into a day out of 365, and a month column to pull the month from the dates.

```

```{r, message=FALSE, warning=FALSE}
all_flower <- all_flower %>%
 mutate(observed_on = parse_date_time(observed_on, orders = c("mdy", "ydm", "dmy"))) %>%
 mutate(year = year(observed_on)) %>%
 mutate(year_day = yday(observed_on)) %>%
 mutate(month = month(observed_on))

```

```

...

```

7. I found the mean date of first sightings of the selected flowers from 2000 to 2014 and from March to June (when these flowers bloom and ruby-throats migrate north) to compare data from the last 10 years to by creating bands of latitude.

```

```{r, message=FALSE, warning=FALSE}
fl_early <- all_flower %>%
  filter(year %in% c(2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012, 2013,
2014) & month %in% c(3, 4, 5, 6))

fl_early_27 <- fl_early %>%
  filter(latitude > 27 & latitude < 33) %>%
  mutate(mean27 = mean(year_day, na.rm = TRUE))

fl_early_33 <- fl_early %>%
  filter(latitude > 33 & latitude < 36) %>%
  mutate(mean33 = mean(year_day, na.rm = TRUE))

fl_early_36 <- fl_early %>%
  filter(latitude > 36 & latitude < 42) %>%
  mutate(mean39 = mean(year_day, na.rm = TRUE))

fl_early_42 <- fl_early %>%
  filter(latitude > 42 & latitude < 45) %>%
  mutate(mean42 = mean(year_day, na.rm = TRUE))

fl_early_45 <- fl_early %>%
  filter(latitude > 45 & latitude < 48) %>%
  mutate(mean42 = mean(year_day, na.rm = TRUE))

fearly1 <- fl_early_27 %>% full_join(fl_early_33)
fearly2 <- fl_early_36 %>% full_join(fl_early_42)
fearly3 <- fearly1 %>% full_join(fl_early_45)

fearly <- fearly2 %>% full_join(fearly3)

flower_early_all <- all_flower %>% full_join(fearly)
...

```

8. I created the dataset to see trends between 2015 and 2025, and in months January to May. I separated sightings into 5 bands of latitude (in increments of approximately 3, excepting 27 to 33 and 36 to 42, since there were so few observations). I rejoined all of these bands into one dataset to plot on a map. I also created a difference column, showing the difference in days of arrival between the new dataset and the 2000 to 2014 average using mean - mean0014.

```

```{r, message=FALSE, warning=FALSE}
flower_new <- flower_early_all %>%

```

```

 filter(year %in% c(2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025) & month
%in% c(3, 4, 5, 6)) %>%
 mutate(mean0014 = case_when(latitude > 27 & latitude < 33 ~ 97.13333
,
 latitude > 33 & latitude < 36 ~ 96.2381,
 latitude > 36 & latitude < 39 ~ 98.94444,
 latitude > 39 & latitude < 42 ~ 98.94444,
 latitude > 42 & latitude < 45 ~ 120.5556,
 latitude > 45 & latitude < 48 ~ 131))

fl_27 <- flower_new %>%
 filter(latitude > 27 & latitude < 33) %>%
 mutate(mean = mean(year_day, na.rm = TRUE)) %>%
 mutate(difference = mean - mean0014)

fl_33 <- flower_new %>%
 filter(latitude > 33 & latitude < 36) %>%
 mutate(mean = mean(year_day, na.rm = TRUE)) %>%
 mutate(difference = mean - mean0014)

fl_36 <- flower_new %>%
 filter(latitude > 36 & latitude < 39) %>%
 mutate(mean = mean(year_day, na.rm = TRUE)) %>%
 mutate(difference = mean - mean0014)

fl_39 <- flower_new %>%
 filter(latitude > 39 & latitude < 42) %>%
 mutate(mean = mean(year_day, na.rm = TRUE)) %>%
 mutate(difference = mean - mean0014)

fl_42 <- flower_new %>%
 filter(latitude > 42 & latitude < 50) %>%
 mutate(mean = mean(year_day, na.rm = TRUE)) %>%
 mutate(difference = mean - mean0014)

fl_1 <- fl_27 %>% full_join(fl_33)
fl_2 <- fl_36 %>% full_join(fl_39)
fl_3 <- fl_42 %>% full_join(fl_2)

fl_4 <- fl_1 %>% full_join(fl_2)

fl_all <- fl_3 %>%
 full_join(fl_4)

...

```

9. I followed the same steps to find data for ruby-throated hummingbird migration and found their average arrival dates for various latitudes between 2000 and 2014. I used data from eBird in addition to iNaturalist and the US National Phenology Network.

My eBird datasets can be found on figshare here:

- 1995-2021: [https://figshare.com/articles/dataset/eBird\\_rthhum\\_2010\\_2014/30535889?file=59303393](https://figshare.com/articles/dataset/eBird_rthhum_2010_2014/30535889?file=59303393)
- 2022-2025: [https://figshare.com/articles/dataset/eBird\\_rthhum\\_2015\\_2025/30535895?file=59303390](https://figshare.com/articles/dataset/eBird_rthhum_2015_2025/30535895?file=59303390)

```
```{r, message=FALSE, warning=FALSE}
```

```
#rth2000 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2000, place_id = 1,
maxresults = 5000)
```

```
...
```

```
```{r, message=FALSE, warning=FALSE, echo=FALSE}
```

```
#rth2001 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2001, place_id = 1,
maxresults = 5000)
```

```
#rth2002 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2002, place_id = 1,
maxresults = 5000)

#rth2003 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2003, place_id = 1,
maxresults = 5000)

#rth2004 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2004, place_id = 1,
maxresults = 5000)

#rth2005 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2005, place_id = 1,
maxresults = 5000)

#rth2006 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2006, place_id = 1,
maxresults = 5000)

#rth2007 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2007, place_id = 1,
maxresults = 5000)

#rth2008 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2008, place_id = 1,
maxresults = 5000)

#rth2009 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2009, place_id = 1,
maxresults = 5000)

#rth2010 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2010, place_id = 1,
maxresults = 5000)

#rth2011 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2011, place_id = 1,
maxresults = 5000)

#rth2012 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2012, place_id = 1,
maxresults = 5000)

#rth2013 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2013, place_id = 1,
maxresults = 5000)

#rth2014 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2014, place_id = 1,
maxresults = 5000)

#rth2015 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2015, place_id = 1,
maxresults = 5000)

#rth2016 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2016, place_id = 1,
maxresults = 5000)

#rth2017 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2017, place_id = 1,
maxresults = 5000)

#rth2018 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2018, place_id = 1,
maxresults = 10000)

#rth2019 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2019, place_id = 1,
maxresults = 10000)

#rth2020 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2020, place_id = 1,
maxresults = 10000)

#rth2021 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2021, place_id = 1,
maxresults = 10000)

#rth2022 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2022, place_id = 1,
maxresults = 10000)

#rth2023 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2023, place_id = 1,
```

```

maxresults = 10000)

#rth2024 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2024, place_id = 1,
maxresults = 10000)

#rth2025 <- get_inat_obs(taxon_name = "Archilochus colubris", year = 2025, place_id = 1,
maxresults = 10000)
```

```{r, warning=FALSE, message=FALSE}
#rth_nat <- rbind(rth2000, rth2001, rth2002, rth2002, rth2003, rth2004, rth2005, rth2005, rth2006,
rth2007, rth2008, rth2009, rth2010, rth2011, rth2012, rth2013, rth2014, rth2015, rth2016, rth2017,
rth2018, rth2019, rth2020, rth2021, rth2022, rth2023, rth2024, rth2025)

#write.csv(rth_nat, "data/rth_nat.csv")

#nnp_download_individual_phenometrics(
 #request_source='Aubrey Monaco',
 #years=c(2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009,
2010,2011,2012,2013,2014,2015,2016,2017,2018,2019,2020,2021,2022,2023, 2024, 2025),
 #species_id=c(352),
 #download_path="data/nnp_rth.csv")

nnp_rth <- read.csv("data/nnp_rth.csv")
nnp_rth <- nnp_rth %>%
 rename(Latitude = latitude)

rth_0014 <- read.csv("data/RTH_1995_2021.csv")
rth_1525 <- read.csv("data/RTH_2022_2025.csv")

rth_ebird <- rth_0014 %>% full_join(rth_1525)

rth_inat <- read.csv("data/rth_nat.csv")
rth_inat <- rth_inat %>%
 rename(Latitude = latitude) %>%
 filter(captive_cultivated == "false" & quality_grade == "research")

rth_some <- rth_ebird %>% full_join(rth_inat)

rth_all <- rth_some %>% full_join(nnp_rth)
```

```{r, message=FALSE, warning=FALSE}
ruby_throat <- rth_all %>%
 mutate(date_new = parse_date_time(Date, orders = c("mdy", "ydm", "dmy"))) %>%
 mutate(year_new = year(date_new)) %>%
 mutate(year_day = yday(date_new))

rth_early <- ruby_throat %>%
 filter(Year %in% c(2000, 2001, 2002, 2003, 2004, 2005, 2006, 2007, 2008, 2009, 2010, 2011, 2012,
2013, 2014) & Month %in% c(3, 4, 5, 6))

rtheearly27 <- rth_early %>%
 filter(Latitude > 27 & Latitude < 33) %>%
 mutate(mean = mean(year_day, na.rm = TRUE))

rtheearly33 <- rth_early %>%
 filter(Latitude > 33 & Latitude < 36) %>%
 mutate(mean = mean(year_day, na.rm = TRUE))

rtheearly36 <- rth_early %>%
 filter(Latitude > 36 & Latitude < 39) %>%
 mutate(mean = mean(year_day, na.rm = TRUE))

```



```

rthearly39 <- rth_early %>%
 filter(Latitude > 39 & Latitude < 45) %>%
 mutate(mean = mean(year_day, na.rm = TRUE))

rthearly45 <- rth_early %>%
 filter(Latitude > 45 & Latitude < 50) %>%
 mutate(mean = mean(year_day, na.rm = TRUE))

rthearly1 <- rthearly27 %>% full_join(rthearly33)
rthearly2 <- rthearly36 %>% full_join(rthearly39)
rthearly3 <- rthearly1 %>% full_join(rthearly45)

rth_early <- rthearly2 %>% full_join(rthearly3)

ruby_throat_all <- ruby_throat %>% full_join(rth_early)

...

```

10. I then similarly created the Ruby-throat dataset for 2015-2025, but I added 3 more bands of latitude to account for the larger number of observations recorded at higher and lower latitudes. I again combined these bands into one dataset, and created a difference column, showing the difference in days of arrival between the new dataset and the 2000 to 2014 average using mean - mean0014.

```

```{r, message=FALSE, warning=FALSE}

rth_all <- ruby_throat_all %>%
  filter(Year %in% c(2015, 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025) & Month
  %in% c(3, 4, 5, 6)) %>%
  mutate(mean0014 = case_when(Latitude > 30 & Latitude < 33 ~ 109.8214,
                              Latitude > 33 & Latitude < 36 ~ 122,
                              Latitude > 36 & Latitude < 39 ~ 141.4,
                              Latitude > 39 & Latitude < 42 ~ 148.7391,
                              Latitude > 42 & Latitude < 45 ~ 148.7391,
                              Latitude > 45 & Latitude < 50 ~ 167.2
                              ))

rth_30 <- rth_all %>%
  filter(Latitude > 30 & Latitude < 36) %>%
  mutate(mean = mean(year_day, na.rm = TRUE)) %>%
  mutate(difference = mean - mean0014)

rth_33 <- rth_all %>%
  filter(Latitude > 33 & Latitude < 36) %>%
  mutate(mean = mean(year_day, na.rm = TRUE)) %>%
  mutate(difference = mean - mean0014)

rth_36 <- rth_all %>%
  filter(Latitude > 36 & Latitude < 39) %>%
  mutate(mean = mean(year_day, na.rm = TRUE)) %>%
  mutate(difference = mean - mean0014)

rth_39 <- rth_all %>%
  filter(Latitude > 39 & Latitude < 42) %>%
  mutate(mean = mean(year_day, na.rm = TRUE)) %>%
  mutate(difference = mean - mean0014)

rth_42 <- rth_all %>%
  filter(Latitude > 42 & Latitude < 45) %>%
  mutate(mean = mean(year_day, na.rm = TRUE)) %>%
  mutate(difference = mean - mean0014)

rth_1 <- rth_30 %>% full_join(rth_33)
rth_2 <- rth_36 %>% full_join(rth_39)
rth_3 <- rth_2 %>% full_join(rth_42)

```

```
rubythroat_all <- rth_1 %>% full_join(rth_3)
```

```
```
```

11. I then created color palettes using viridis scales and colorNumeric from the Leaflet package for my two maps, specifying to use the difference column in each dataset to map the colors to.

```
```{r}
```

```
cpal <- colorNumeric(palette = "mako", domain = fl_all$difference, na.color = NA)
```

```
rpal <- colorNumeric(palette = "mako", domain = rubythroat_all$difference, na.color = NA)
```

```
```
```

```
Download and clean all required data
```

To download Ruby-throated Hummingbird data from eBird on your own, visit:

<https://media.ebird.org/catalog?birdOnly=true&taxonCode=rthhum>, filter by year (I did this in two separate actions [1995-2021 and 2022-2025] to ensure I could download all of the data), and filter by location (United States)

```
Results
```

```
\\~200 words\\
```

Tables and figures (maps and other graphics) are carefully planned to convey the results of your analysis. Intense exploration and evidence of many trials and failures. The author looked at the data in many different ways before coming to the final presentation of the data.

```
Show tables, plots, etc. and describe them.r
```

```
Trends in Flower Blooms (2015 - 2025)
```

```
```{r}
```

```
leaflet(fl_all) %>%
```

```
  addProviderTiles(provider = "Esri") %>%
```

```
  addSimpleGraticule(interval = 3) %>%
```

```
  addCircleMarkers(~longitude, ~ latitude, radius = 1, color = ~cpal(difference)) %>%
```

```
  addLegend(pal = cpal, values = ~difference, title = "Difference From Early Average (days)",  
position = "bottomleft")
```

```
```
```

```
Trends in Ruby-Throated Hummingbird Sightings (2015 - 2025)
```

```
```{r}
```

```
leaflet(rubythroat_all) %>%
```

```
  addProviderTiles(provider = "Esri") %>%
```

```
  addSimpleGraticule(interval = 3) %>%
```

```
  addCircleMarkers(~Longitude, ~ Latitude, radius = 1, color = ~rpal(difference)) %>%
```

```
  addLegend(pal = rpal, values = ~difference, title = "Difference From Early Average (days)",  
position = "bottomleft")
```

```
```
```

```
Conclusions
```

```
\\~200 words\\
```

Trends for the preferred flowers have shown that its timing of first observations have changed from the 2000-2014 average, and at most latitudes, when moving higher, there is a pattern of earlier sightings each year. But for the lowest latitude, flowers have been sighted earlier than some higher latitudes. This may be due to hummingbirds remaining at these lower latitudes longer.

Similarly, the average first observation date for Ruby-throated hummingbirds has been shifting earlier with increasing latitudes. But similarly to the flower trends, the observations at lower latitudes are happening later on average.

#### # References

All sources are cited in a consistent manner

Thanks for checking out my web site!